

Abstract Algebra II

Homework 6

Due: 13th May 2026

Submit only Questions 1 and 2, i.e., everything.

1. Let K be a field of characteristic zero, and $f \in K[x]$ be a quartic with distinct roots u_1, u_2, u_3, u_4 in some splitting field L . Set

$$\alpha = u_1u_2 + u_3u_4, \quad \beta = u_1u_3 + u_2u_4, \quad \gamma = u_1u_4 + u_2u_3.$$

Compute the structure of $\text{Aut}(K(\alpha, \beta, \gamma)/K)$.

2. Let $P(t) = t^4 - 30t^2 + 1$.
 - (a) Show that $P(t)$ is irreducible over $\mathbb{Q}$.
 - (b) Show that $P(t) \bmod p \in \mathbb{F}_p[t]$ is reducible for all prime numbers p .
 - (c) Let K denote the splitting field of $P(t)$ over $\mathbb{Q}$. Find $\text{Gal}(K/\mathbb{Q})$ and all the intermediate fields of the extension $K/\mathbb{Q}$.